

Chapter 4 – Linux Terminal Commands

Introduction

Linux is one of the most important operating systems in:

- Ethical Hacking
- Cybersecurity
- Servers
- Cloud Computing
- Networking

Cybersecurity professionals spend a huge amount of time working in Linux terminals.

So learning Linux commands is extremely important.

1.File & Directory Management Commands

These commands help manage:

- Files
 - Folders
 - Directories
 - Data organization
-

i) Is Command

Meaning

LS = List Directory Contents

Used to display files and folders inside a directory.

Syntax

ls

Example

ls

Output

Documents notes.txt song.mp3

Why We Use It?

To check:

- Which files exist
 - Which folders exist
 - Available content in current directory
-

ii) cd Command

Meaning

CD = Change Directory

Used to move from one folder to another.

Syntax

cd foldername

Example

cd Documents

Moves into the Documents folder.

Linux is Case Sensitive

cd Documents

 **Correct**

cd documents

 **Wrong** (if actual folder starts with capital D)

iii) pwd Command

Meaning

PWD = Print Working Directory

Shows current directory location.

Syntax

pwd

Example Output

/home/admin/Desktop

iv) mkdir Command

Meaning

MKDIR = Make Directory

Creates a new folder.

Syntax

mkdir foldername

Example

```
mkdir hacking_notes
```

Creates a folder named:

```
hacking_notes
```

v) rmdir Command

Meaning

Remove Empty Directory

Deletes empty folders only.

Syntax

```
rmdir foldername
```

Example

```
rmdir testfolder
```

Important Note

rmdir works only if folder is empty.

vi) touch Command

Purpose

Creates an empty file.

Syntax

```
touch filename
```

Example

`touch notes.txt`

Creates an empty text file.

vii) rm Command

Meaning

RM = Remove

Deletes files or directories.

Syntax

`rm filename`

Example

`rm notes.txt`

Deletes the file.

Force Delete

Syntax

`rm -rf foldername`

Meaning

Option Purpose

-r Recursive delete

-f Force delete

Important Warning

This command permanently deletes files/folders.

No recycle bin recovery usually exists.

viii) cp Command

Meaning

CP = Copy

Copies files and folders.

Syntax

cp source destination

Example

cp notes.txt backup.txt

Copies notes.txt into backup.txt.

ix) mv Command

Meaning

MV = Move

Used to:

- Move files
 - Rename files
-

Move File Example

mv notes.txt /home/admin/Documents

Rename File Example

```
mv old.txt new.txt
```

x) find Command

Purpose

Searches for files and directories.

Syntax

```
find location -name filename
```

Example

```
find /home -name notes.txt
```

Searches for notes.txt inside /home.

2.File Viewing & Editing Commands

These commands help:

- Read files
 - Edit files
 - Analyze logs
 - View configurations
-

i) cat Command

Purpose

Displays complete file content.

Syntax

cat filename

Example

cat notes.txt

ii) more Command

Purpose

Views file content page-by-page.

Example

more logfile.txt

Useful for large files.

iii) less Command

Purpose

Advanced version of more.

Allows scrolling:

- Up
 - Down
 - Search inside file
-

Example

less logfile.txt

Why less is Better?

You can navigate easily inside huge log files.

Cybersecurity analysts often use it for:

- System logs
 - Apache logs
 - Authentication logs
-

iv) head Command

Purpose

Shows first 10 lines of a file.

Example

head logfile.txt

v) tail Command

Purpose

Shows last 10 lines of a file.

Example

tail logfile.txt

Cybersecurity Importance of tail

Used for monitoring live logs.

Example

Admins monitor:

- Failed logins
- Server attacks
- Authentication attempts

using tail command.

vi) nano Text Editor

Definition

Simple terminal-based text editor.

Easy for beginners.

Example

nano notes.txt

Save & Exit

Shortcut Purpose

CTRL + O Save

CTRL + X Exit

Example Workflow

- 1 Open file using nano
 - 2 Write text
 - 3 Press CTRL + O to save
 - 4 Press CTRL + X to exit
-

⚡ vii) vi / vim Editor

📘 Definition

Advanced Linux text editor.

Widely used by:

- Linux admins
- Ethical hackers
- Developers

📌 Why Vim is Popular?

- ✓ Fast
- ✓ Lightweight
- ✓ Powerful
- ✓ Works on almost every Linux system

🔒 Linux File Permissions

📘 Why Permissions Matter?

Permissions decide:

- ✓ Who can read files
- ✓ Who can modify files
- ✓ Who can execute files

This is extremely important in cybersecurity.

Permission Values

Permission Value

Read (r) 4

Write (w) 2

Execute (x) 1

i) Viewing Permissions

Command

ls -l

Example Output

-rw-r--r-- 1 admin admin 120 notes.txt

Understanding Permission Structure

-rw-r--r--

First Character

Symbol Meaning

- File

d Directory

Three Permission Slots

Slot	Represents
------	------------

First	User/Owner
-------	------------

Second Group	
--------------	--

Third	Others
-------	--------

Example Breakdown

-rw-r--r--

User Permissions

rw-

User can:

- ☒ Read
- ☒ Write
- ☒ Execute

Group Permissions

r--

Group can:

- ☒ Read only

Others Permissions

r--

Other users can:

- ☒ Read only
-

Numeric Permission Calculation

Example

Suppose:

User needs:

- Read
- Write
- Execute

Calculation:

$$4 + 2 + 1 = 7$$

Group Needs

- Read
- Write

$$4 + 2 = 6$$

Others Need

- Read only

$$4$$

Final Permission

$$764$$

ii) chmod Command

Meaning

CHMOD = Change Mode

Changes file permissions.

Syntax

chmod permissions filename

Example

chmod 764 wordlist.txt

Meaning of 764

User Type Permission

User	rwX
Group	rw-
Others	r--

iii) chown Command

Meaning

Change file owner.

Example

chown admin notes.txt

Changes file owner to admin.

iv) chgrp Command

Meaning

Change group ownership.

Example

chgrp developers notes.txt

v) umask Command

Purpose

Sets default permissions for new files/folders.

Example

umask 022

3.User Management Commands

i) whoami

Purpose

Shows currently logged-in user.

Example

whoami

 ii) id

 Purpose

Displays:

- User ID (UID)
 - Group ID (GID)
-

 Example

id

 iii) adduser / useradd

 Purpose

Creates a new user.

 Example

sudo adduser hacker

 iv) passwd

 Purpose

Changes user password.

 Example

passwd hacker

✖ v) deluser / userdel

📘 Purpose

Deletes a user.

⚙ Example

sudo deluser hacker

🔄 vi) su

📘 Meaning

Switch User.

⚙ Example

su root

👑 vii) sudo

📘 Purpose

Runs command with administrator privileges.

⚙ Example

sudo apt update

👥 viii) groups

📘 Purpose

Shows user group memberships.

✖ Example

groups

📦 **4.Package Management Commands (Debian/Kali Linux)**

Kali Linux is Debian-based.

Package management is done using:

📦 **APT**

(Advanced Package Tool)

🔄 i) **apt update**

📘 **Purpose**

Updates package list information.

✖ Example

sudo apt update

🧠 **What It Actually Does?**

It refreshes repository information and checks:

- Which packages are available
 - Latest versions
 - Security updates
-

⬆ ii) **apt upgrade**

📘 **Purpose**

Upgrades installed packages.

Example

sudo apt upgrade

Difference Between Update & Upgrade

Command	Purpose
---------	---------

apt update	Refresh package database
------------	--------------------------

apt upgrade	Install updated versions
-------------	--------------------------

Daily Life Example

Action	Real Life Example
--------	-------------------

Update	Checking app store for updates
--------	--------------------------------

Upgrade	Actually installing updates
---------	-----------------------------

iii) apt install

Purpose

Installs software package.

Example

sudo apt install nmap

Installs Nmap tool.

iv) apt remove

Purpose

Removes installed package.

Example

```
sudo apt remove nmap
```

v) dpkg -i

Purpose

Installs .deb package files manually.

Example

```
sudo dpkg -i software.deb
```

vi) apt-cache search

Purpose

Searches packages.

Example

```
apt-cache search wireshark
```



5. System Monitoring & Information Commands

These commands monitor:

- CPU
- RAM
- Processes
- Disk
- System performance



i) top



Purpose

Shows running processes in real time.



Example

top



ii) htop



Purpose

Improved version of top.

More user-friendly.



Example

htop



Important Note

Usually needs installation first.

 iii) free -h

Purpose

Displays RAM and swap memory usage.

Example

free -h

What is Swap Memory?

Swap memory uses hard disk space as temporary RAM.

Windows calls similar feature:

Virtual Memory

Why Swap/Virtual Memory is Used?

If RAM becomes full:

System temporarily uses hard disk space.

Important Note

Hard disk is much slower than RAM.

So excessive swapping slows system performance.

Windows Virtual Memory Setup

Command

sysdm.cpl

Steps

- 1 Press Windows + R
 - 2 Run sysdm.cpl
 - 3 Open Advanced Tab
 - 4 Performance Settings
 - 5 Advanced → Virtual Memory
 - 6 Choose drive
 - 7 Set custom size in MB
 - 8 Restart system
-

iv) df -h

Purpose

Shows disk space usage.

Example

df -h

v) du -sh

Purpose

Shows folder size.

Example

du -sh Downloads

vi) uptime

Purpose

Shows how long system has been running.

Example

uptime

vii) uname -a

Purpose

Displays detailed system information.

Example

uname -a

viii) lscpu

Purpose

Shows CPU details.

Example

lscpu

ix) lsblk

Purpose

Displays connected storage devices.

 Example

lsblk

 x) ps aux

 Purpose

Shows running processes.

 Example

ps aux

 6. Networking Commands

Networking is extremely important in cybersecurity.

 i) ping

 Purpose

Checks network connectivity.

 Example

ping google.com

 What Ping Does?

Sends ICMP packets to target.

Checks:

- Connectivity

- Response time
 - Packet loss
-

ii) ifconfig / ip a

Purpose

Displays IP address information.

Examples

ifconfig

or

ip a

Shows

- IP Address
 - MAC Address
 - Network Interface
-

iii) netstat

Purpose

Shows:

- Open ports
 - Listening services
 - Active connections
-

Linux Example

netstat -tulnp

Windows Example

netstat -a

Why Important in Cybersecurity?

Helps identify:

- Open ports
 - Suspicious services
 - Malware connections
-

iv) traceroute

Purpose

Traces network path between source and destination.

Linux Example

traceroute google.com

Windows Equivalent

tracert google.com

Why Traceroute is Useful?

Helps identify:

- Network delays
 - Routing problems
 - ISP issues
-

Quick Revision Table

Topic	Key Command
List files	ls
Change folder	cd
Create folder	mkdir
Remove file	rm
File permissions	chmod
Current user	whoami
Install package	apt install
View RAM	free -h
View IP	ip a
Check ports	netstat
Network testing	ping
